

1 Method Description

1.1 Derivation of Main Equations

By performing a force balance (so assuming no acceleration) on the 2D body we get that

$$-\sigma_{xx}A_x + \left(\sigma_{xx} + \frac{\partial\sigma_{xx}}{\partial x}\Delta x\right)A_x - \sigma_{yx}A_y + \left(\sigma_{yx} + \frac{\partial\sigma_{yx}}{\partial y}\Delta y\right)A_y + S_x = 0 \quad (1)$$

and

$$-\sigma_{yy}A_y + \left(\sigma_{yy} + \frac{\partial\sigma_{yy}}{\partial y}\Delta y\right)A_y - \sigma_{xy}A_x + \left(\sigma_{xy} + \frac{\partial\sigma_{xy}}{\partial x}\Delta x\right)A_x + S_y = 0. \quad (2)$$

Note that for example A_x is the area of the face in the y-z plane. Its normal is in the x-direction. Here, S is the internal force. By canceling out some like terms we have

$$\frac{\partial\sigma_{xx}}{\partial x}\Delta xA_x + \frac{\partial\sigma_{yx}}{\partial y}\Delta yA_y + S_x = 0 \quad (3)$$

and

$$\frac{\partial\sigma_{yy}}{\partial y}\Delta yA_y + \frac{\partial\sigma_{xy}}{\partial x}\Delta xA_x + S_y = 0. \quad (4)$$

Next, if we assume a rectangular control volume of dimensions $\Delta x \times \Delta y \times \Delta z$, ΔyA_y and ΔxA_x are both equivalent to V . S then becomes f , which represents the force per unit volume in that direction. Also, the shear stresses can be written with τ .

$$\frac{\partial\sigma_{xx}}{\partial x} + \frac{\partial\tau_{xy}}{\partial y} + f_x = 0 \quad (5)$$

$$\frac{\partial\tau_{xy}}{\partial x} + \frac{\partial\sigma_{yy}}{\partial y} + f_y = 0, \quad (6)$$

or

$$\nabla \cdot \boldsymbol{\sigma} = -\mathbf{f}, \quad (7)$$

where

$$\boldsymbol{\sigma} = \begin{bmatrix} \sigma_{xx} & \tau_{xy} \\ \tau_{xy} & \sigma_{yy} \end{bmatrix}. \quad (8)$$

It has been assumed that $\tau_{xy} = \tau_{yx}$ due to an angular moment balance. These equations, when written in matrix form are

$$\begin{bmatrix} \frac{\partial}{\partial x} & 0 & \frac{\partial}{\partial y} \\ 0 & \frac{\partial}{\partial y} & \frac{\partial}{\partial x} \end{bmatrix} \begin{bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \tau_{xy} \end{bmatrix} = - \begin{bmatrix} f_x \\ f_y \end{bmatrix} \quad (9)$$

or simply

$$B^T \boldsymbol{\sigma} = -\mathbf{f}, \quad (10)$$

where B is a differential operator that relates stresses to force equilibrium. It is assumed that displacements are sufficiently small that equilibrium conditions that were established for the initial geometry remain valid. Note that the stresses here are total stresses. In the case where we have pore pressure, we replace total stress with the definition of effective stress, $\boldsymbol{\sigma} = \boldsymbol{\sigma}' + \alpha p \mathbf{I}$, with $\mathbf{I}$ being the identity matrix and α the Biot coefficient, such that our equation becomes

$$\nabla \cdot (\boldsymbol{\sigma}' + \alpha p \mathbf{I}) = -\mathbf{f}. \quad (11)$$

Here, compression is taken as positive. This results in

$$\nabla \cdot \boldsymbol{\sigma}' + \nabla (\alpha p) = -\mathbf{f}, \quad (12)$$

or

$$B^T \boldsymbol{\sigma}' + \nabla (\alpha p) = -\mathbf{f}, \quad (13)$$

where

$$\nabla (\alpha p) = \alpha \begin{bmatrix} \frac{\partial p}{\partial x} \\ \frac{\partial p}{\partial y} \end{bmatrix}, \quad (14)$$

for a spatially-constant α .

1.2 Hooke's Law

Hooke's Law reads,

$$\sigma'_{ij} = \lambda \epsilon_{kk} \delta_{ij} + 2\mu \epsilon_{ij}, \quad (15)$$

where σ' is an effective stress. $\lambda = \frac{E\nu}{(1+\nu)(1-2\nu)}$, so

$$\sigma'_{ij} = \frac{E\nu}{(1+\nu)(1-2\nu)} \epsilon_{kk} \delta_{ij} + 2\mu \epsilon_{ij}, \quad (16)$$

and $\mu = \frac{E}{2(1+\nu)}$, so

$$\sigma'_{ij} = \frac{E\nu}{(1+\nu)(1-2\nu)} \epsilon_{kk} \delta_{ij} + \frac{E}{(1+\nu)} \epsilon_{ij}. \quad (17)$$

Since $\epsilon_{kk} = \epsilon_{xx} + \epsilon_{yy}$ in Cartesian coordinates for plane strain in x-y plane ($\epsilon_{zz} = 0$),

$$\sigma'_{xx} = \frac{E}{(1+\nu)(1-2\nu)} ((1-\nu)\epsilon_{xx} + \nu\epsilon_{yy}), \quad (18)$$

$$\sigma'_{yy} = \frac{E}{(1+\nu)(1-2\nu)} ((1-\nu)\epsilon_{yy} + \nu\epsilon_{xx}), \quad (19)$$

and

$$\tau_{xy} = \frac{E}{2(1+\nu)} \gamma_{xy}. \quad (20)$$

Note that the shear strain here is the engineering shear strain, such that $\gamma_{xy} = 2\epsilon_{xy}$. These equations can be written in matrix form as:

$$\begin{bmatrix} \sigma'_{xx} \\ \sigma'_{yy} \\ \tau_{xy} \end{bmatrix} = \frac{E}{(1+\nu)(1-2\nu)} \begin{bmatrix} 1-\nu & \nu & 0 \\ \nu & 1-\nu & 0 \\ 0 & 0 & \frac{1-2\nu}{2} \end{bmatrix} \begin{bmatrix} \epsilon_{xx} \\ \epsilon_{yy} \\ \gamma_{xy} \end{bmatrix}, \quad (21)$$

which can also be written as

$$\boldsymbol{\sigma}' = D\boldsymbol{\epsilon}, \quad (22)$$

where

$$\boldsymbol{\sigma}' = \begin{bmatrix} \sigma'_{xx} \\ \sigma'_{yy} \\ \tau_{xy} \end{bmatrix}, \quad \boldsymbol{\epsilon} = \begin{bmatrix} \epsilon_{xx} \\ \epsilon_{yy} \\ \gamma_{xy} \end{bmatrix}. \quad (23)$$

Although not solved for explicitly, the out-of-plane effective stress is given by $\sigma'_{zz} = \frac{E\nu}{(1+\nu)(1-2\nu)} (\epsilon_{xx} + \epsilon_{yy})$.

1.3 Rewriting the Strains

The definition for ϵ under the assumption of small strains is

$$\epsilon = \frac{\Delta L}{L}. \quad (24)$$

The limit as the length of our system goes to zero causes

$$\lim_{L \rightarrow 0} \frac{\Delta L}{L} = \frac{du}{dx}. \quad (25)$$

This can be done for both directions such that

$$\epsilon_{xx} = \frac{\partial u_x}{\partial x} \quad (26)$$

and

$$\epsilon_{yy} = \frac{\partial u_y}{\partial y}. \quad (27)$$

The engineering shear strain, $\gamma_{ij} = 2\epsilon_{ij}$, for small displacement gradients is given by

$$\gamma_{xy} = \frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x}. \quad (28)$$

These equations can then be written in matrix form yielding

$$\begin{bmatrix} \epsilon_{xx} \\ \epsilon_{yy} \\ \gamma_{xy} \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial x} & 0 \\ 0 & \frac{\partial}{\partial y} \\ \frac{\partial}{\partial y} & \frac{\partial}{\partial x} \end{bmatrix} \begin{bmatrix} u_x \\ u_y \end{bmatrix} \quad (29)$$

or simply

$$\tilde{\epsilon} = B\tilde{u} \quad (30)$$

and also

$$\epsilon = Bu \quad (31)$$

Terms with a tilde are continuous whereas when displacement and strain are written without a tilde it means that they are nodal values.

Setting Up The System Of Equations

By combining Equations 10, 22, and 31 we get

$$B^T DBu = -f - \alpha \nabla p \quad (32)$$

Basis Functions

The displacement, $\tilde{u}$, can be found at any point in the element through the use of the values at the nodes and basis functions such that

$$\tilde{u} = N_1 u_1 + N_2 u_2 + N_3 u_3 + N_4 u_4 \quad (33)$$

for an element with four points or, more generally in matrix form,

$$\tilde{u} = Nu. \quad (34)$$

Here, N represents the basis functions and u represents the value of displacement (in both x and y) at the nodes. For an element with four nodes, this looks like

$$\tilde{u} = \begin{bmatrix} N_1 & 0 & N_2 & 0 & N_3 & 0 & N_4 & 0 \\ 0 & N_1 & 0 & N_2 & 0 & N_3 & 0 & N_4 \end{bmatrix} \begin{bmatrix} u_{x1} \\ u_{y1} \\ u_{x2} \\ u_{y2} \\ u_{x3} \\ u_{y3} \\ u_{x4} \\ u_{y4} \end{bmatrix} \quad (35)$$

For a square element with linear basis functions with the node numbering beginning in the upper left and moving horizontally before vertically, the basis functions are

$$N_1 = \left(1 - \frac{x}{a}\right) \left(1 - \frac{y}{b}\right) \quad (36)$$

$$N_2 = \frac{x}{a} \left(1 - \frac{y}{b}\right) \quad (37)$$

$$N_3 = \left(1 - \frac{x}{a}\right) \frac{y}{b} \quad (38)$$

$$N_4 = \frac{x}{a} \frac{y}{b} \quad (39)$$

Here, a is the length of the element in the x direction and b is the length of the element in the y direction. Now, Equation 32 which we had previously is written like

$$B^T D B N u = -f - \nabla p \quad (40)$$

1.4 Including Body Forces and Pore Pressures

Multiplying the above result by the transpose of the basis functions yields:

$$N^T B^T D B N u = -N^T f - N^T \nabla p \quad (41)$$

Recall that f is a stress per length. This multiplied by the basis functions is a stress per length in a given direction at a given location. The integral of this over x and y is the force per length (where the length is into the plane) at any given location (matching the location of the displacement which will be at the nodes). The basis function for pore pressure is just one everywhere because the pore pressure is constant in each element. Pore pressure is also turned into a distributed load, then multiplied by the length of the element side along which it is applied. This yields a force per length (recalling that we are in plane strain and the length is into the plane).

$$\int_0^a \int_0^b N^T B^T D B N u \partial \xi \partial \eta = - \int_0^a \int_0^b N^T f \partial \xi \partial \eta - \int_0^a \int_0^b N^T \nabla p \partial \xi \partial \eta \quad (42)$$

$$\int_0^a \int_0^b N^T B^T D B N u \partial \xi \partial \eta = F + P \quad (43)$$

Here, P is basically the pressure of each element times the length of the side of the element in the direction in which the force is being applied times the Biot coefficient. F is similar. As an example, see the image below.

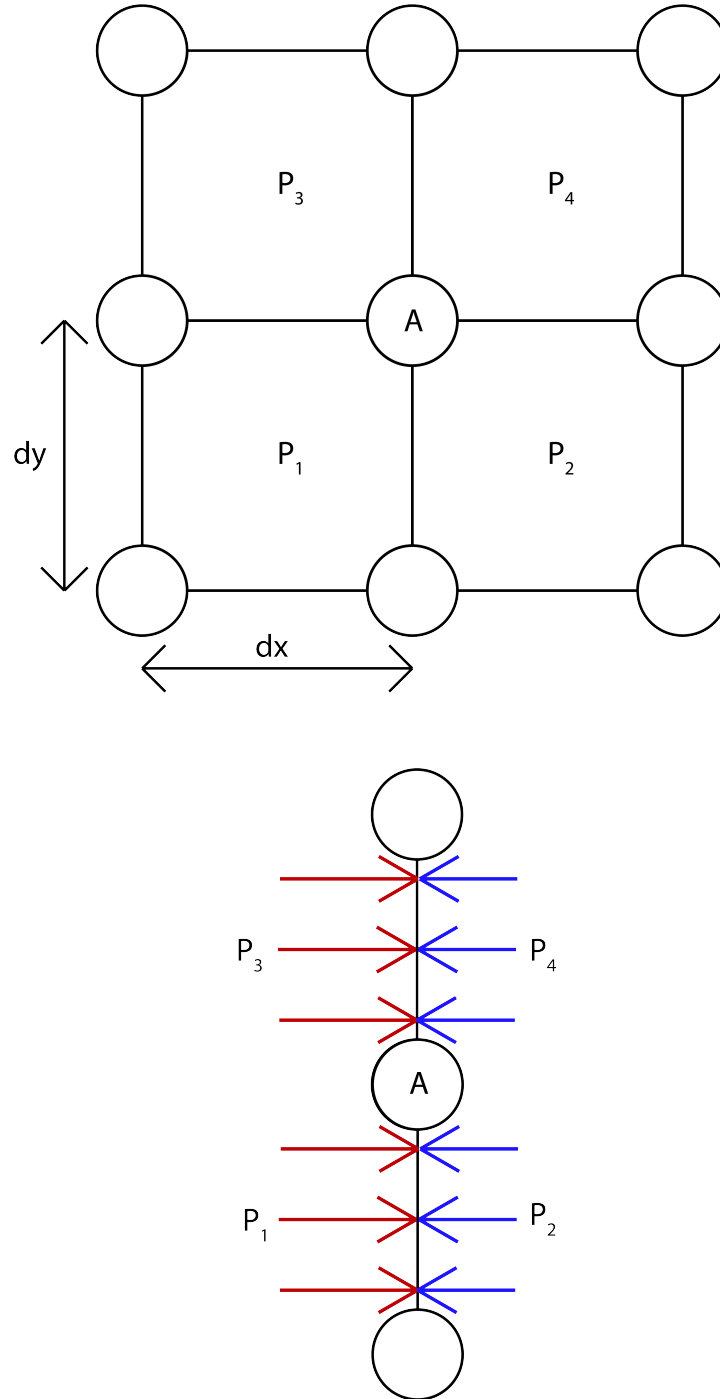


Figure 1: How pore pressures are applied

The force due to pore pressure, P_{Ax} , on node A in the x-direction would then be give by

$$P_{Ax} = \alpha \frac{dy}{2} (P_1 + P_3 - P_2 - P_4) \quad (44)$$

The same principle is applied for adding distributed loads. Body forces (which again in plane strain are actually a force per length where the length is going into the plane) are simply applied as a two single values, one for the x and one for the y directions.

1.5 Setting Up The Final Equation

Also recall that this has been done for one element. These integrals can now be performed for an element by evaluating a stiffness matrix, k_m which is defined as

$$k_m = \int_0^a \int_0^b N^T B^T D B N \partial\xi \partial\eta \quad (45)$$

This will yield an 8x8 matrix (for quadrilateral elements). The definitions of N , B , and D (for plane strain) are listed below and the values of each element of the resulting matrix are shown in the table (for quadrilateral elements and linear basis functions). This can also be done with Gauss Integration.

$$D = \frac{E}{(1+\nu)(1-2\nu)} \begin{bmatrix} 1-\nu & \nu & 0 \\ \nu & 1-\nu & 0 \\ 0 & 0 & \frac{1-2\nu}{2} \end{bmatrix} \quad (46)$$

$$B N = \begin{bmatrix} \frac{\partial N_1}{\partial x} & 0 & \frac{\partial N_2}{\partial x} & 0 & \frac{\partial N_3}{\partial x} & 0 & \frac{\partial N_4}{\partial x} & 0 \\ 0 & \frac{\partial N_1}{\partial y} & 0 & \frac{\partial N_2}{\partial y} & 0 & \frac{\partial N_3}{\partial y} & 0 & \frac{\partial N_4}{\partial y} \\ \frac{\partial N_1}{\partial y} & \frac{\partial N_1}{\partial x} & \frac{\partial N_2}{\partial y} & \frac{\partial N_2}{\partial x} & \frac{\partial N_3}{\partial y} & \frac{\partial N_3}{\partial x} & \frac{\partial N_4}{\partial y} & \frac{\partial N_4}{\partial x} \end{bmatrix} \quad (47)$$

Realize that $N^T B^T = (B N)^T$. Now, for example, $k_m(1,1)$ can be found as:

$$k_m(1,1) = \frac{E}{(1+\nu)(1-2\nu)} \int_0^a \int_0^b \left[(1-\nu) \left(\frac{\partial N_1}{\partial x} \right)^2 + \frac{1-2\nu}{2} \left(\frac{\partial N_1}{\partial y} \right)^2 \right] \partial\xi \partial\eta \quad (48)$$

where the basis functions can be directly plugged in and integrated. This can usually not be done analytically and typically Gauss quadrature is used to solve the integrals and build the stiffness matrix. It is possible to take each derivative manually and fill up the local (and ultimately the global) stiffness matrix in this way. The required integrals for this are listed in the Appendix. Note that the displacements weren't involved in the integral because these are displacements at the node locations and these values are only at the nodes. Their integral over an area is therefore a constant.

Now the equation looks like

$$k_m u = (F + P) \quad (49)$$

and can be solved for displacements

$$u = k_m^{-1} (F + P) \quad (50)$$

1.6 Solving For Strains

Recall that

$$\epsilon = -Bu. \quad (51)$$

However, what we actually do is

$$\epsilon = BNu. \quad (52)$$

so that we can solve for strains at any location in the element by plugging in different values of the shape functions into the shape functions.

So, to solve for strains all we need to do is build a B matrix. This can be done by solving for the Jacobian, which is shown below:

$$J = \begin{bmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial y}{\partial \xi} \\ \frac{\partial x}{\partial \eta} & \frac{\partial y}{\partial \eta} \end{bmatrix}. \quad (53)$$

These values can be found by averaging the difference in x-locations between nodes (ie the x distances along the lines connecting the nodes) and dividing by the the length of the local coordinate.

Next, the derivatives required for the B matrix are found by taking the inverse of the Jacobian and multiplying by the derivative of the shape functions in the local coordinates. For example,

$$\frac{\partial N_1}{\partial x} = \frac{\partial N_1}{\partial \xi} \frac{\partial \xi}{\partial x}. \quad (54)$$

Then, the BN matrix is created by placing the correct derivatives in the correct place such that

$$BN = \begin{bmatrix} \frac{\partial N_1}{\partial x} & 0 & \frac{\partial N_2}{\partial x} & 0 & \frac{\partial N_3}{\partial x} & 0 & \frac{\partial N_4}{\partial x} & 0 \\ 0 & \frac{\partial N_1}{\partial y} & 0 & \frac{\partial N_2}{\partial y} & 0 & \frac{\partial N_3}{\partial y} & 0 & \frac{\partial N_4}{\partial y} \\ \frac{\partial N_1}{\partial y} & \frac{\partial N_1}{\partial x} & \frac{\partial N_2}{\partial y} & \frac{\partial N_2}{\partial x} & \frac{\partial N_3}{\partial y} & \frac{\partial N_3}{\partial x} & \frac{\partial N_4}{\partial y} & \frac{\partial N_4}{\partial x} \end{bmatrix}. \quad (55)$$

Finally, the strains can be solved for using

$$\epsilon = BNu. \quad (56)$$

Again, the strains are evaluated at the location corresponding to the local coordinates used in N .

1.7 Solving For Stresses

The stresses can then simply be solved for in the following manner:

$$\sigma' = D\epsilon \quad (57)$$

and total stress can be found with

$$\sigma = \sigma' + p \quad (58)$$

2 Developing The Code

To begin with, it is very important to properly keep track of which element is which and which node is which. Globally, the elements will be numbered as can be seen in Figure ??.

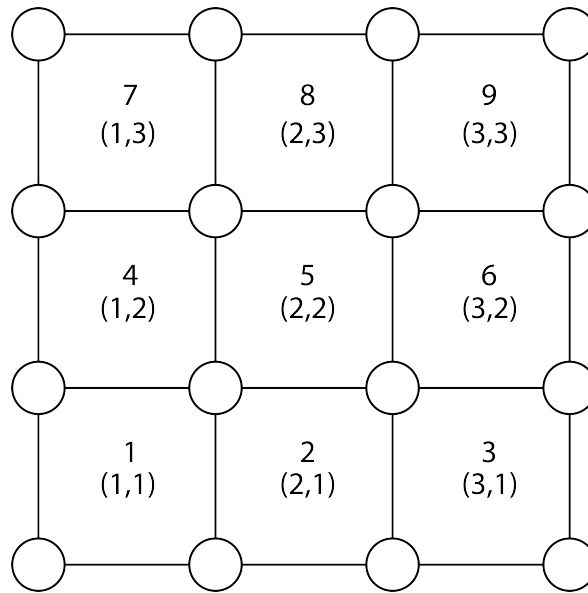


Figure 2: Global Element Numbering. The element number can be seen for when the elements are in vector format and the matrix location of each element can be seen below that for when the elements need to be in matrix format.

The node numbering for the same system shown in Figure ?? can be seen in Figure ??.

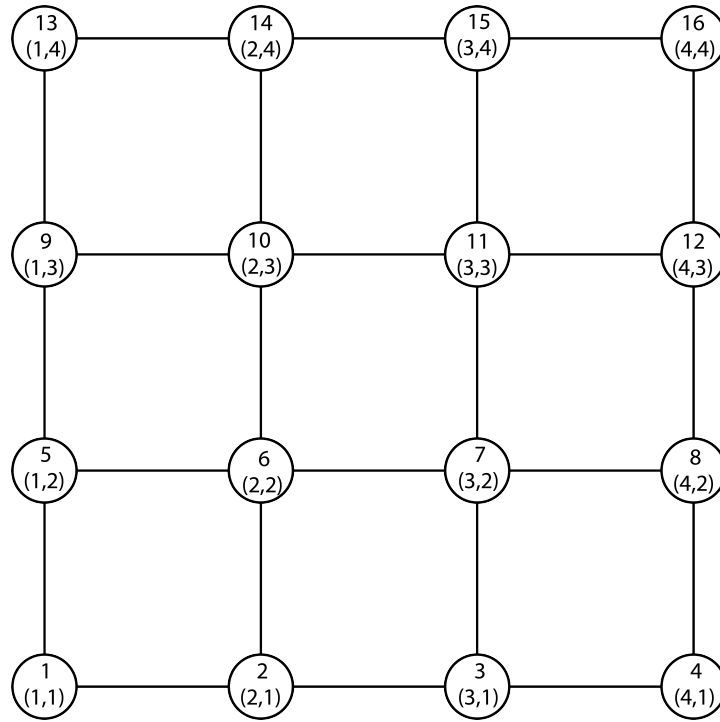


Figure 3: Global Nodal Numbering. The node number can be seen for when the nodes are in vector format and the matrix location of each element can be seen below that for when the nodes need to be in matrix format.

Properties should be defined at the elements, not at the nodes. Additionally, here, the center of the FVM cell will be located at the center of the element. The FVM cells will be the same size as the elements and will also be quadrilateral. From here, it is recommended that a new function be created for each of the following subsections.

2.1 Building the local stiffness matrix

The terms of the local stiffness matrix are given by

$$k_m = \int_0^a \int_0^b N^T B^T D B N \partial \xi \partial \eta. \quad (59)$$

2.2 Applying the local stiffness matrix to the global stiffness matrix

2.3 Applying external forces

2.4 Applying pore pressure

2.5 Applying boundary conditions

After the stiffness matrix and force vector have been created, the easiest way to fix displacements on a given node is to set the row of the stiffness matrix corresponding to the node of interest to be all zeros. Be aware that each node corresponds to two rows, one for the x direction one for the y direction. Then, on the same row, the element corresponding to the diagonal (i.e. where the column number is the same as the row number) for that row should be set to be equal to one. Now, this entire row is full

of zeros except for a one on the diagonal. If both directions for this node are to be fixed, this must be done for each row.

Next, the row of the force vector corresponding to the node (and direction) in question should be set to be equal to the desired displacement. If the node is to be a fixed displacement, this value will be a zero.

2.6 Solving for strains

Ultimately, the goal of this section will be to evaluate

$$\epsilon = BNu. \quad (60)$$

for strains by creating the matrices B , N , and vector u . To do this, first, two matrices should be built containing the x and y displacements of the nodes for each element, El_{dx} and El_{dy} . These should be of size $[N_e, 4]$, with each row representing an element and each column representing either the x or y displacement of the nodes of that element. Care should be taken to put the displacements in the correct order based on the number shown in Figure ??, with regard to which column the nodal displacements are placed in.

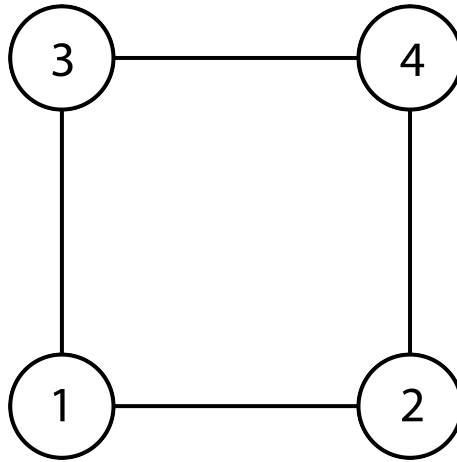


Figure 4: Local Node Numbering

Next, the same process should be performed to create two more matrices, Loc_x and Loc_y , one for the global x-location of the nodes and one for the global y-location of the nodes. Again these matrices should be of size $[N_e, 4]$ and should contain the the locations of the nodes belonging to the element of that particular row. This operation and the previous operation which collected the x and y displacements of each node could, for example, be performed in the same for loop (looping over elements).

At this stage, another for loop over the elements can be begun (of course this can be combined with the first for loop if you'd like). Next, the global coordinates of each of the nodes corresponding to this element should be collected in a new variable, $coord$, which is of size $[4, 2]$ as we have quadrilateral elements and two displacement directions. The rows should be in the order of the local node numbering (seen in Figure ??) and the columns should be x and then y. These values can be taken from the recently created Loc_x and Loc_y variables. These values are the global coordinates of the nodes. The resulting matrix is

$$coord = \begin{bmatrix} x_1 & y_1 \\ x_2 & y_2 \\ x_3 & y_3 \\ x_4 & y_4 \end{bmatrix}. \quad (61)$$

At this stage, the displacements of the nodes corresponding to the current element should be collected in a new variable, $disp$, which is of size $[1, 8]$. The odd-numbered columns should contain the x displacements (in local node order) and the even-numbered columns the y displacements (in local node order) i.e.,

$$disp = [u_{x1} \ u_{y1} \ u_{x2} \ u_{y2} \ u_{x3} \ u_{y3} \ u_{x4} \ u_{y4}]. \quad (62)$$

These values can be taken from the El_{dx} and El_{dy} variables. This variable, $disp$, corresponds to the variable u from Equation 56.

Now it is important to recall our local directions and shape functions. Our local directions are setup such that ξ increases in the x-direction and η increases in the y-direction. Both local variables begin in the center of the given element. Each element has a length of 2 in both the local directions (even if globally the element is much bigger or smaller than this). This means, for example that local node 1 is located at $(\xi = -1, \eta = -1)$ and local node 4 is located at $(\xi = 1, \eta = 1)$. This can be seen in Figure 5.

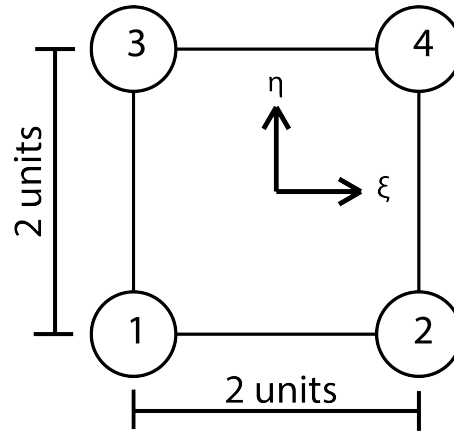


Figure 5: Local Coordinates

The shape functions (in this case linear), N , are therefore defined as

$$\begin{bmatrix} N_1 \\ N_2 \\ N_3 \\ N_4 \end{bmatrix} = \begin{bmatrix} \frac{1}{4} (1 - \xi) (1 - \eta) \\ \frac{1}{4} (1 + \xi) (1 - \eta) \\ \frac{1}{4} (1 - \xi) (1 + \eta) \\ \frac{1}{4} (1 + \xi) (1 + \eta) \end{bmatrix}. \quad (63)$$

We can determine at which point we want to evaluate our shape functions (and soon the derivatives of the shape functions and ultimately the strains and stresses themselves) by defining ξ and η . The

simplest case is to define them both as zero, the middle of the element. Then, we can create a new matrix containing the derivatives of the shape functions with respect to the local directions, i.e.,

$$dN = \begin{bmatrix} \frac{\partial N_1}{\partial \xi} & \frac{\partial N_2}{\partial \xi} & \frac{\partial N_3}{\partial \xi} & \frac{\partial N_4}{\partial \xi} \\ \frac{\partial N_1}{\partial \eta} & \frac{\partial N_2}{\partial \eta} & \frac{\partial N_3}{\partial \eta} & \frac{\partial N_4}{\partial \eta} \end{bmatrix} = \begin{bmatrix} -\frac{1}{4}(1-\eta) & \frac{1}{4}(1-\eta) & -\frac{1}{4}(1+\eta) & \frac{1}{4}(1+\eta) \\ -\frac{1}{4}(1-\xi) & -\frac{1}{4}(1+\xi) & \frac{1}{4}(1-\xi) & \frac{1}{4}(1+\xi) \end{bmatrix}. \quad (64)$$

Now the Jacobian, J , which is defined as

$$J = \begin{bmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial y}{\partial \xi} \\ \frac{\partial x}{\partial \eta} & \frac{\partial y}{\partial \eta} \end{bmatrix}, \quad (65)$$

can be calculated as

$$J = dN * coord. \quad (66)$$

To see why this is the case, we can begin with

$$\tilde{x} = \sum N x, \quad (67)$$

where $\tilde{x}$ is a location within the element (determined by the local coordinates used in the basis functions, N) and x is the nodal location. The summation is over all nodes of that element. N effectively maps the local coordinates to the global coordinates. By taking the derivative of $\tilde{x}$ with respect to each of the local variables, we have, for example,

$$\frac{\partial \tilde{x}}{\partial \xi} = \sum \frac{\partial N}{\partial \xi} x. \quad (68)$$

Note that the nodal locations, x , are independent of the local variables. When written out for both x and y , the above equation becomes

$$\begin{bmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial y}{\partial \xi} \end{bmatrix} = \begin{bmatrix} \frac{\partial N_1}{\partial \xi} & \frac{\partial N_2}{\partial \xi} & \frac{\partial N_3}{\partial \xi} & \frac{\partial N_4}{\partial \xi} \end{bmatrix} \begin{bmatrix} x_1 & y_1 \\ x_2 & y_2 \\ x_3 & y_3 \\ x_4 & y_4 \end{bmatrix}. \quad (69)$$

When this is done for both local variables we get

$$\begin{bmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial y}{\partial \xi} \\ \frac{\partial x}{\partial \eta} & \frac{\partial y}{\partial \eta} \end{bmatrix} = \begin{bmatrix} \frac{\partial N_1}{\partial \xi} & \frac{\partial N_2}{\partial \xi} & \frac{\partial N_3}{\partial \xi} & \frac{\partial N_4}{\partial \xi} \\ \frac{\partial N_1}{\partial \eta} & \frac{\partial N_2}{\partial \eta} & \frac{\partial N_3}{\partial \eta} & \frac{\partial N_4}{\partial \eta} \end{bmatrix} \begin{bmatrix} x_1 & y_1 \\ x_2 & y_2 \\ x_3 & y_3 \\ x_4 & y_4 \end{bmatrix}, \quad (70)$$

which is equivalent to

$$J = dN * coord. \quad (71)$$

Now, the derivatives of the basis functions with respect to the global variables can be found with,

$$deriv = J^{-1}dN. \quad (72)$$

This results in

$$deriv = \begin{bmatrix} \frac{\partial N_1}{\partial x} & \frac{\partial N_2}{\partial x} & \frac{\partial N_3}{\partial x} & \frac{\partial N_4}{\partial x} \\ \frac{\partial N_1}{\partial y} & \frac{\partial N_2}{\partial y} & \frac{\partial N_3}{\partial y} & \frac{\partial N_4}{\partial y} \end{bmatrix} \quad (73)$$

These derivatives can then be used to fill up the BN matrix because this matrix is equivalent to

$$BN = \begin{bmatrix} \frac{\partial N_1}{\partial x} & 0 & \frac{\partial N_2}{\partial x} & 0 & \frac{\partial N_3}{\partial x} & 0 & \frac{\partial N_4}{\partial x} & 0 \\ 0 & \frac{\partial N_1}{\partial y} & 0 & \frac{\partial N_2}{\partial y} & 0 & \frac{\partial N_3}{\partial y} & 0 & \frac{\partial N_4}{\partial y} \\ \frac{\partial N_1}{\partial y} & \frac{\partial N_1}{\partial x} & \frac{\partial N_2}{\partial y} & \frac{\partial N_2}{\partial x} & \frac{\partial N_3}{\partial y} & \frac{\partial N_3}{\partial x} & \frac{\partial N_4}{\partial y} & \frac{\partial N_4}{\partial x} \end{bmatrix}. \quad (74)$$

Finally, strains can be calculated using

$$\epsilon = BN * u^T. \quad (75)$$

Note that these strains are the strains of each element located at the location within the element given by the values chosen for ξ and η . Also, when done in this way u^T is the transpose of the u vector used above. Finally, the strains, given by ϵ , will be of size $[3, 1]$, where the first row will be ϵ_{xx} , the second row ϵ_{yy} , and the third row ϵ_{xy} . As we are still in the for loop (over each element), this process should be repeated (starting back from) for each element. These strains can then be stored in a larger matrix of size $[3, 1]$. Note that here the strains are positive in dilation (and the stresses calculated from them will be positive in tension).

Solving for stresses

In order to solve for stresses, to begin with, the matrix D should be defined for the plane strain case,

$$D = \frac{E}{(1 + \nu)(1 - 2\nu)} \begin{bmatrix} 1 - \nu & \nu & 0 \\ \nu & 1 - \nu & 0 \\ 0 & 0 & \frac{1 - 2\nu}{2} \end{bmatrix}. \quad (76)$$

This matrix changes for each element, so this process should be repeated for each element. Ideally, we perform this calculation (and the calculation of stress shown below) right after calculating strains. Note that properties are constant within one element (as they are defined per element).

The stresses can then be readily solved with

$$\sigma = D * \epsilon. \quad (77)$$

Here again the stresses are found in a column vector of size $[3, 1]$, where the first column is σ_{xx} , the second row is σ_{yy} , and the third row is σ_{xy} . This process should be repeated for each element.

3 Validation

3.1 Validation Using Boussinesq

Boussinesq presented analytical solutions for point loads. Helwany 2007 used these solutions for the validation of an FEM simulator. One such problem he presented is below.

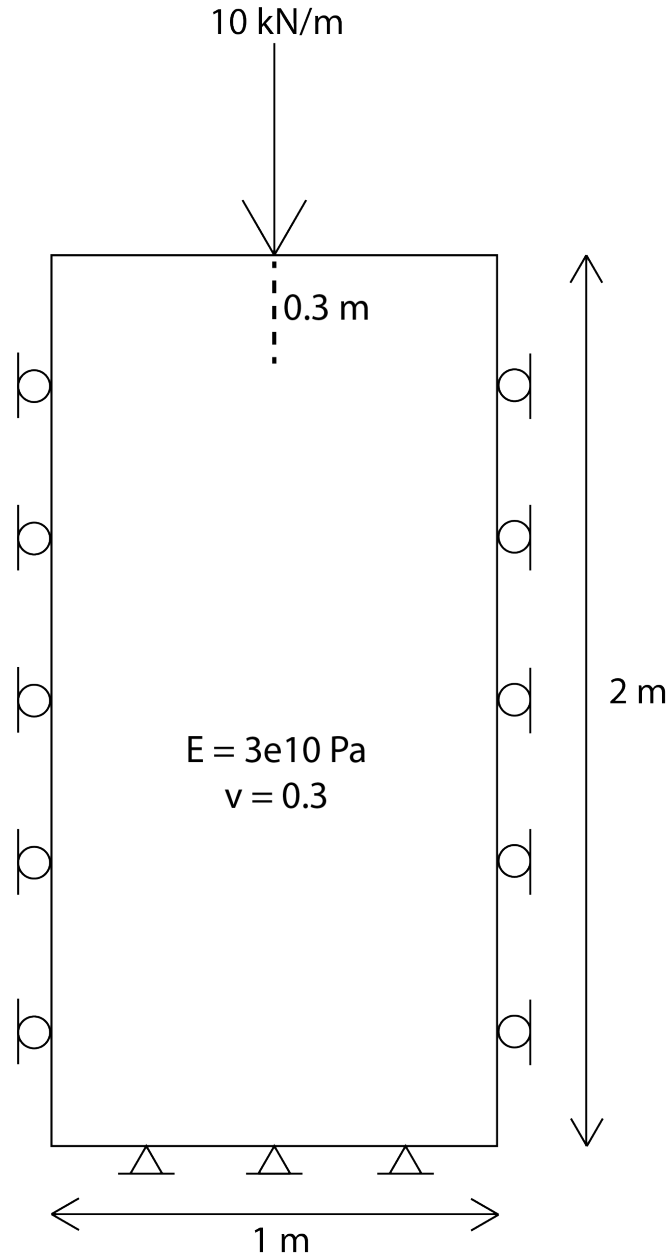


Figure 6: Validation Against Boussinesq Analytical Solution Setup

The analytical Boussinesq 1883 solution for plane strain due to a line load is:

$$\Delta\sigma_z = \frac{2qz^3}{\pi(x^2 + z^2)^2} \quad (78)$$

which reduces to

$$\Delta\sigma_z = \frac{2q}{\pi \cdot z} \quad (79)$$

if we set x to be equal to zero and just look at how stress changes with depth. This was then compared against my FEM solution and the results are below.

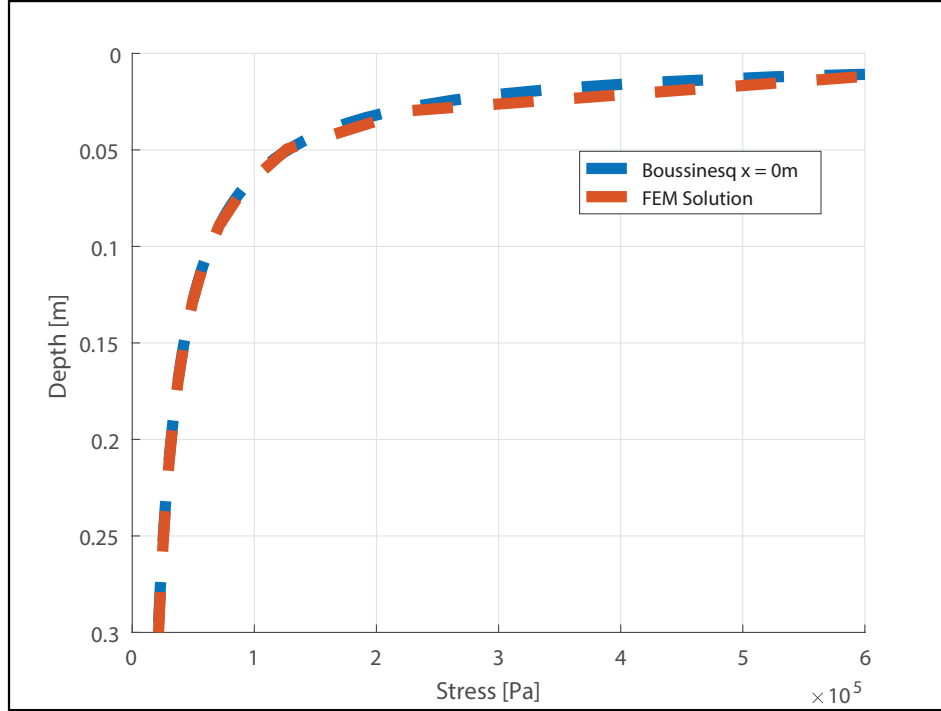


Figure 7: Validation Against Boussinesq Analytical Solution

3.2 Validation Using Definition of Plane Strain

By applying a load across the top of $-1000 \left[\frac{N}{m} \right]$ (which corresponds to placing a force of $-1000 \cdot \frac{L_x}{N_x} [N]$ on each node in the middle of the system and half this force on the edge nodes), we expect a vertical stress of $-1000 [Pa]$ throughout the system. The horizontal and shear stresses should be zero.

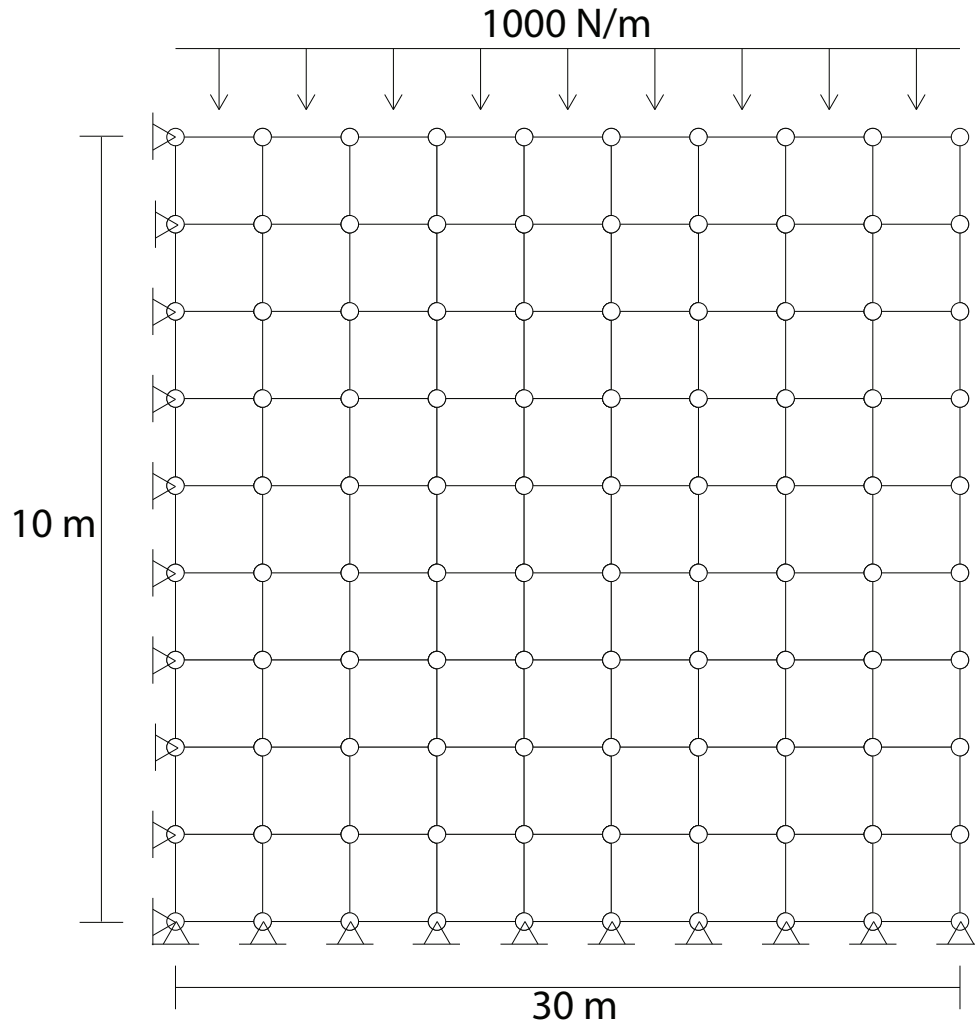


Figure 8: Setup

Next, if we take a look at the plane strain formulation,

$$\begin{bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \tau_{xy} \end{bmatrix} = \frac{-E}{(1+\nu)(1-2\nu)} \begin{bmatrix} 1-\nu & \nu & 0 \\ \nu & 1-\nu & 0 \\ 0 & 0 & \frac{1-2\nu}{2} \end{bmatrix} \begin{bmatrix} \epsilon_{xx} \\ \epsilon_{yy} \\ \gamma_{xy} \end{bmatrix}, \quad (80)$$

we can plug in values for a given problem. For example, if the force was applied such that the stress equals $-1000 [Pa]$, Young's Modulus equals $1 \times 10^9 [Pa]$, and Poisson's ratio equals 0.3, we get the following system of equations:

$$\begin{bmatrix} 0 \\ -1000 \\ 0 \end{bmatrix} = \frac{-1 \times 10^9}{(1.3)(0.4)} \begin{bmatrix} 0.7 & .3 & 0 \\ 0.3 & 0.7 & 0 \\ 0 & 0 & 0.2 \end{bmatrix} \begin{bmatrix} \epsilon_{xx} \\ \epsilon_{yy} \\ \gamma_{xy} \end{bmatrix}, \quad (81)$$

or

$$0 = -1.9231 \times 10^9 (0.7 \cdot \epsilon_{xx} + 0.3 \epsilon_{yy}) \quad (82)$$

$$-1000 = -1.9231 \times 10^9 (0.3 \cdot \epsilon_{xx} + 0.7 \epsilon_{yy}) \quad (83)$$

$$0 = -1.9231 \times 10^9 \cdot 0.2 \cdot \epsilon_{xy}. \quad (84)$$

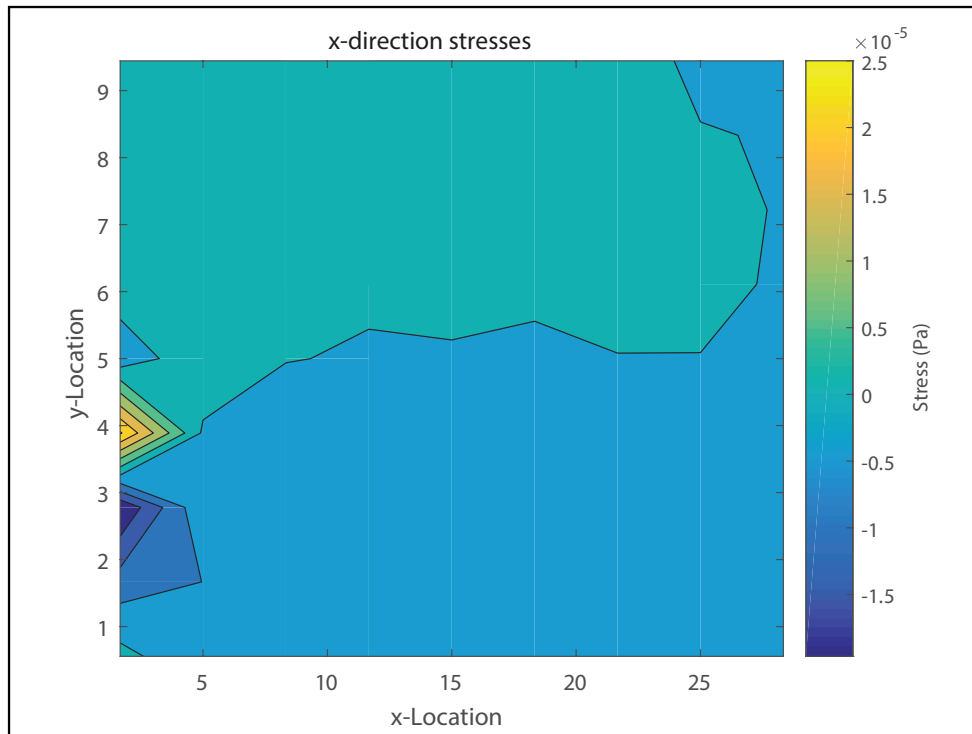


Figure 9: X-Stresses

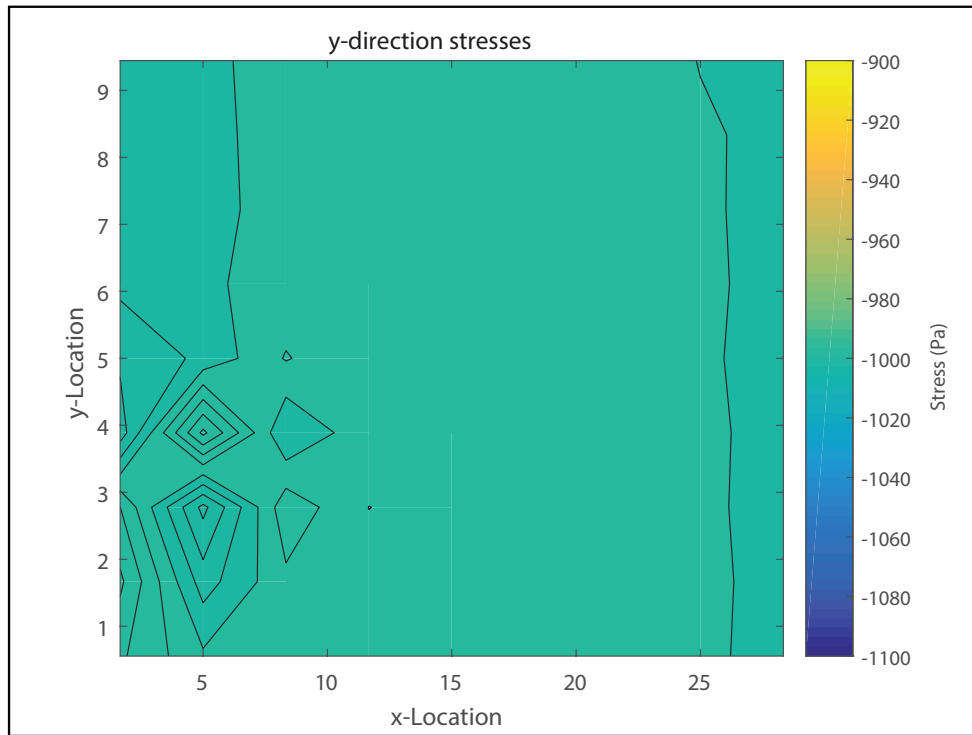


Figure 10: Y-Stresses

We could, in theory, solve for the expected strains. Alternatively, we could run the problem in our simulator and simply plug in our values of strain and see if they satisfy the equations. This is what I did. I found that $\epsilon_{xx} = 3.9 \times 10^{-7}$ and $\epsilon_{yy} = -9.1 \times 10^{-7}$. These values satisfy the equations.

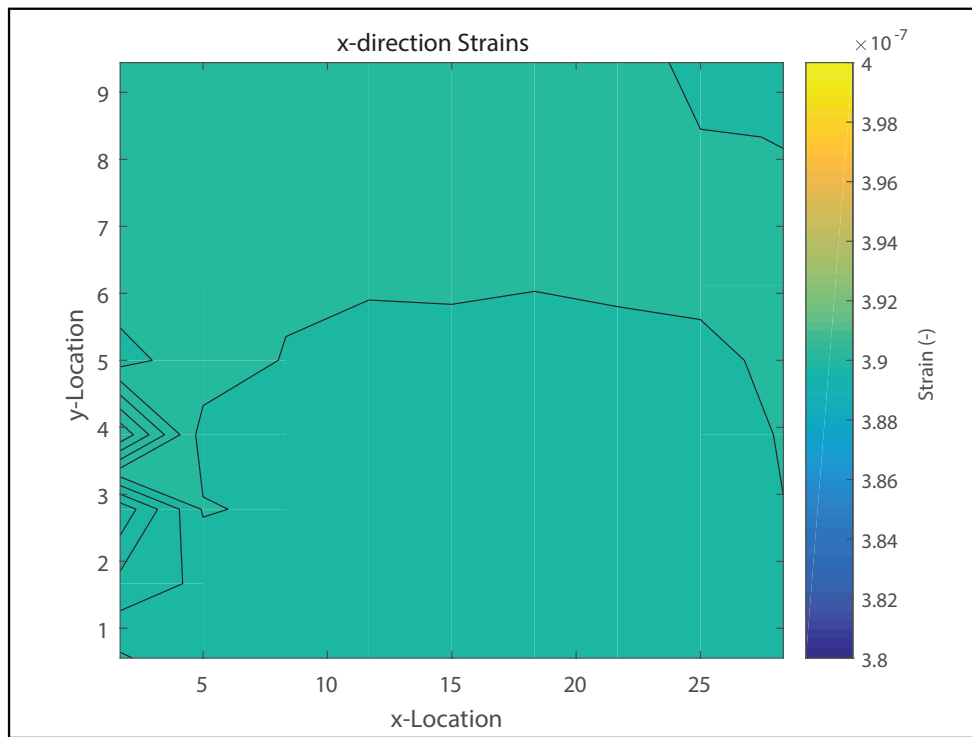


Figure 11: X-Strains

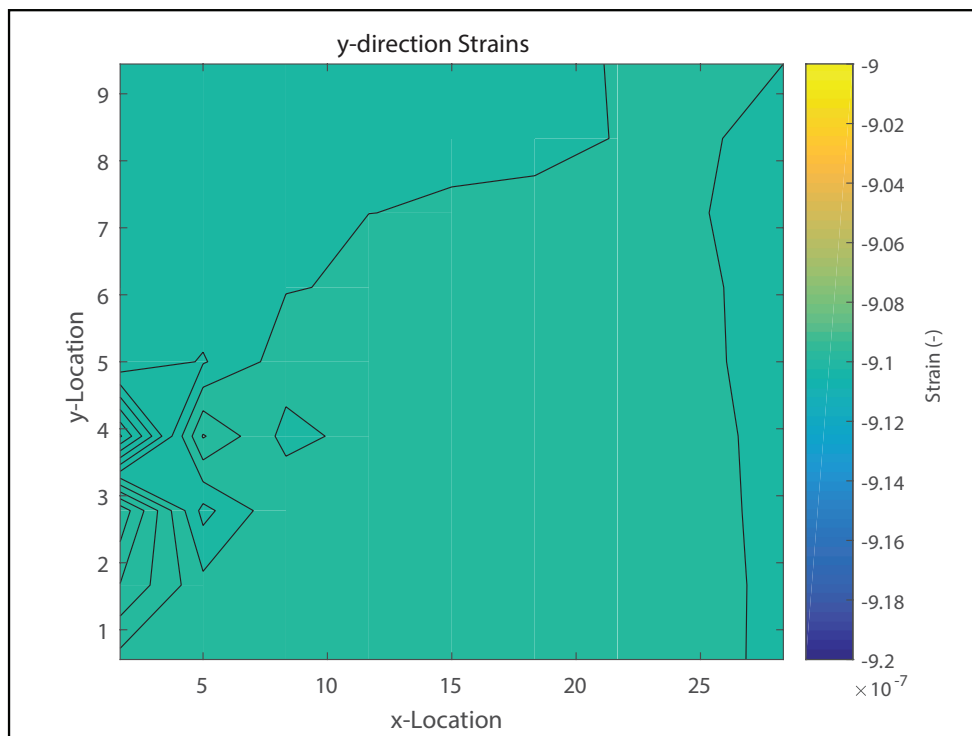


Figure 12: Y-Strains

Also we can note that the maximum displacement in the y-direction (occurring where the forces are applied) is equal to the strain multiplied by the length in the y-direction. The same check holds for the

x-displacement at the right-hand edge.

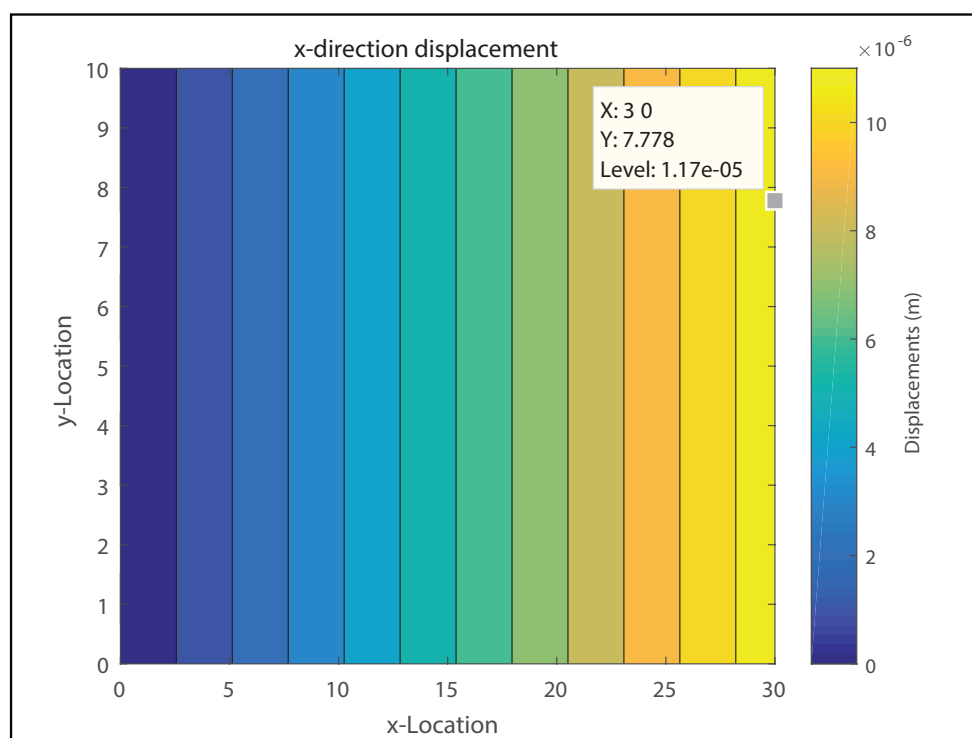


Figure 13: X-Displacements

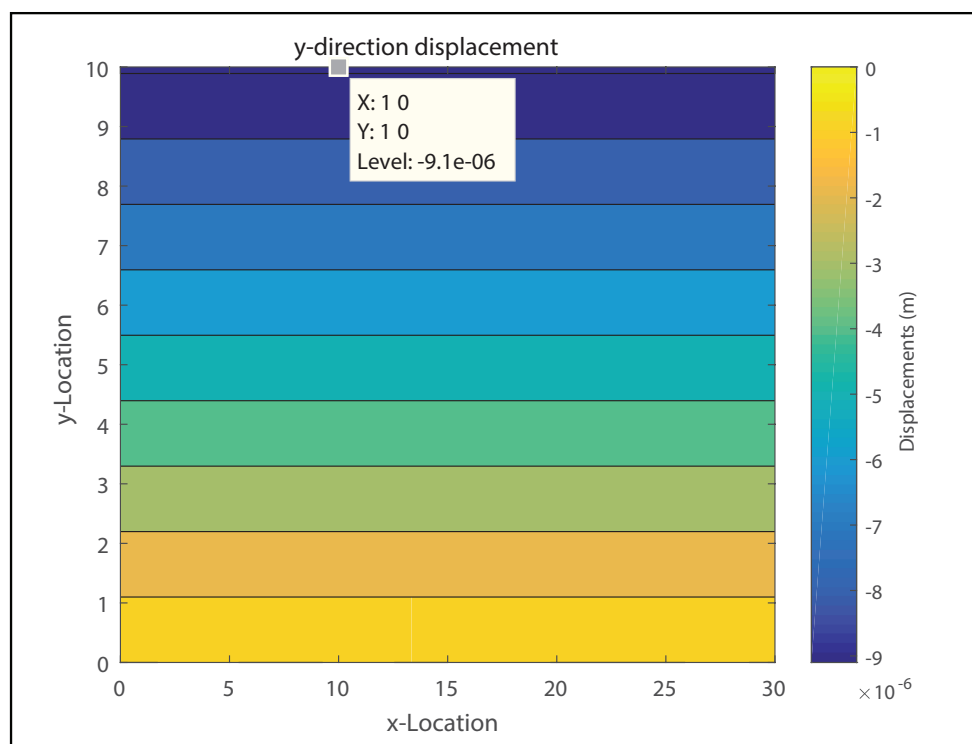


Figure 14: Y-Displacements

3.3 Validation Using ABAQUS

Next, the problem depicted below was run both in ABAQUS and my simulator. As it is difficult to depict the results in a comparative manner in 2-D, a cut was made along the red dashed line in the figure below and stresses were analyzed along this line for both the ABAQUS results and mine.

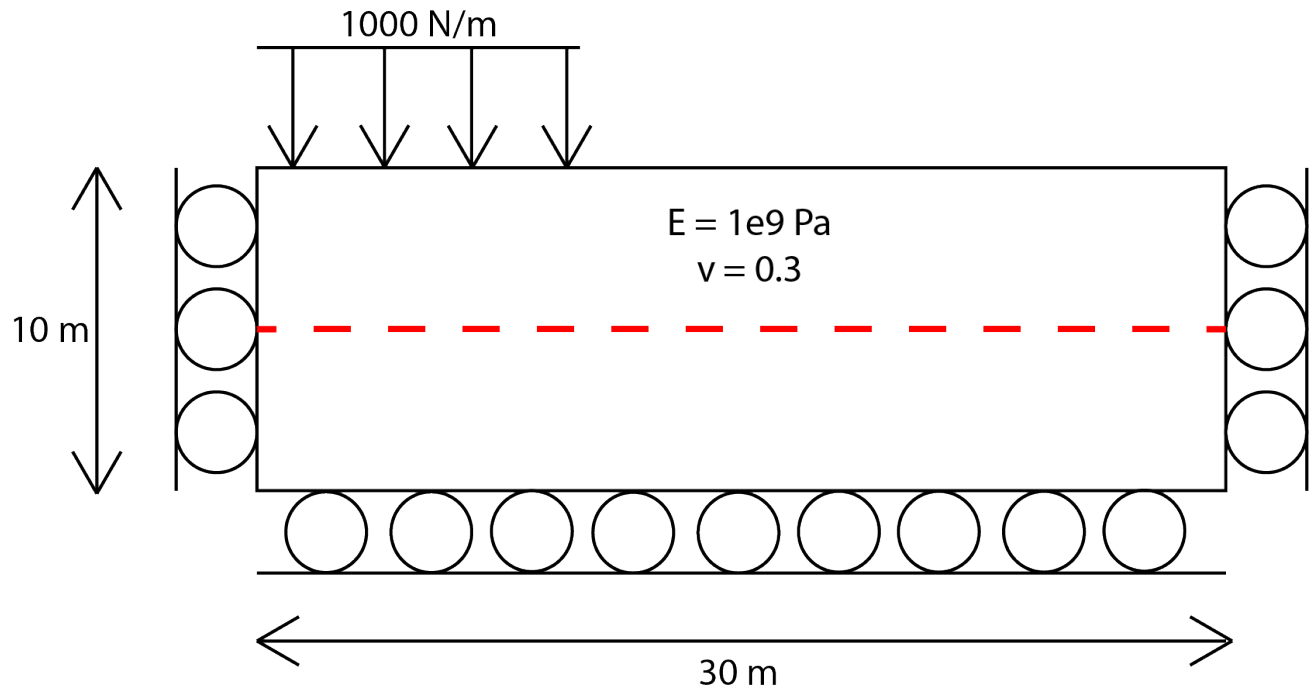
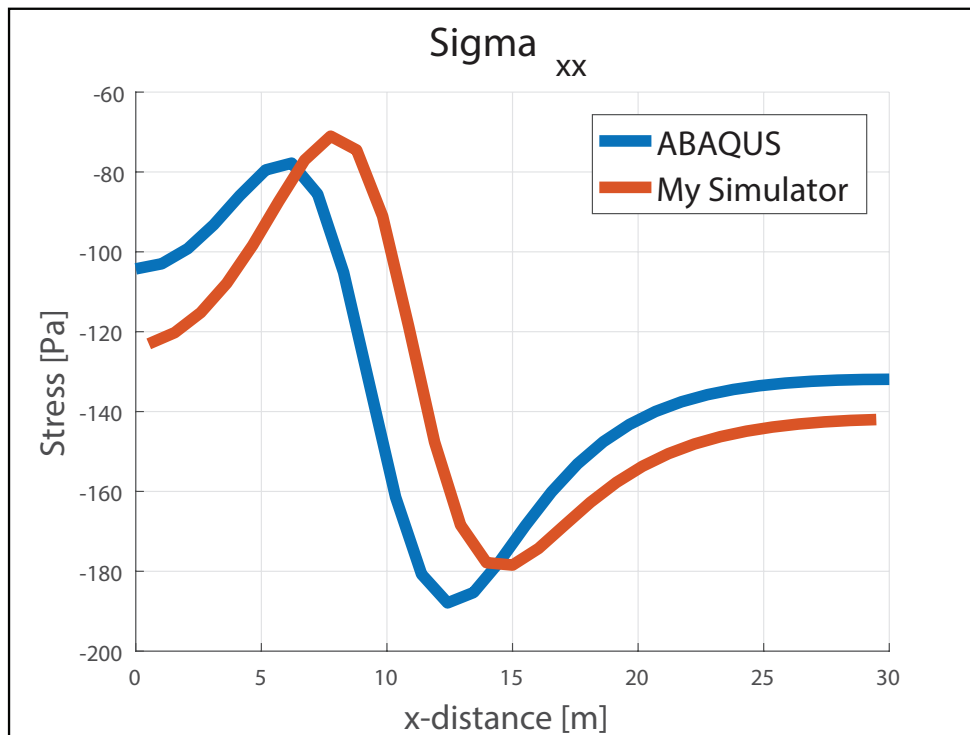
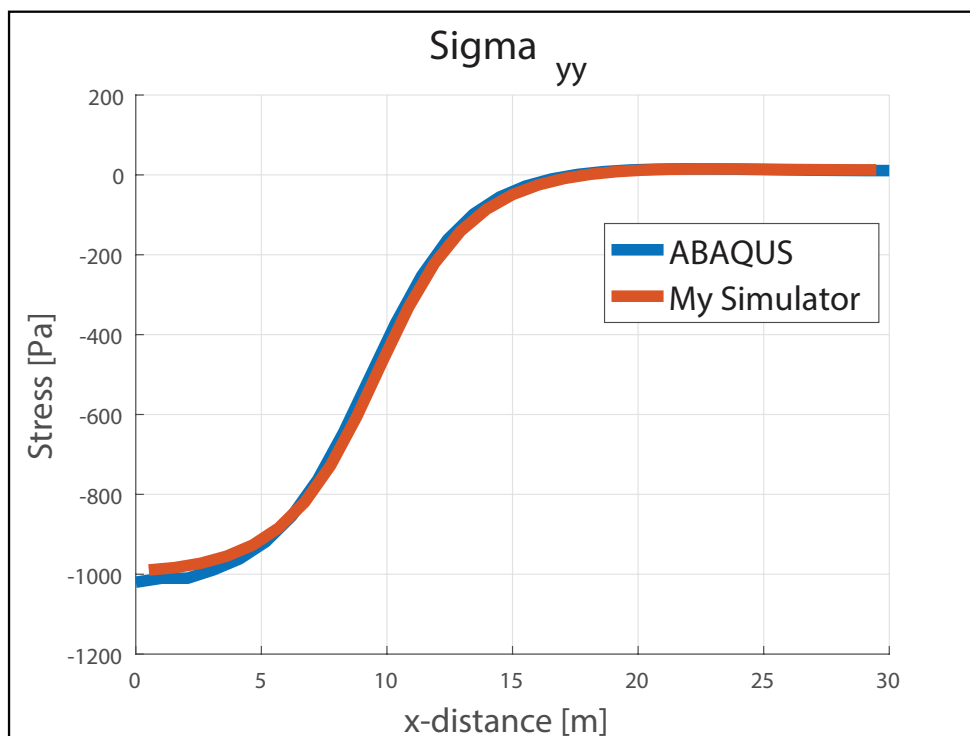
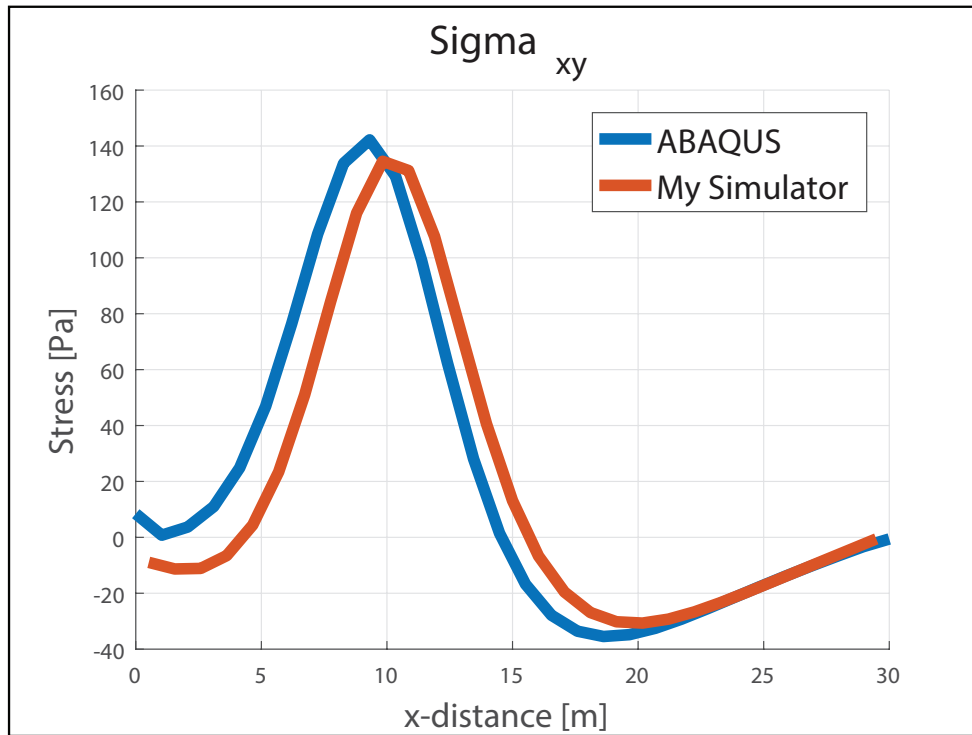


Figure 15: FEM Setup

Figure 16: Comparison of σ_{xx} Figure 17: Comparison of σ_{yy}

Figure 18: Comparison of σ_{xy}

3.4 Poroelastic Validation - Terzaghi

Terzaghi's consolidation problem consists of a confined sample (on three sides) filled with water. The sample is assumed to be in plane strain conditions. A distributed load, q , in $[Pa]$ is applied to the top of the sample. Unlike Terzaghi's original work, the analytical solutions taken from ? and ? will be compressible. A constant pressure of zero (atmospheric) is assumed at the top of the sample; no flow boundaries are defined for the other three edges. Additionally, these edges are assumed to be fixed and are defined in my model to have zero displacement. The top edge is allowed to move freely. A schematic can be seen in Figure ??.

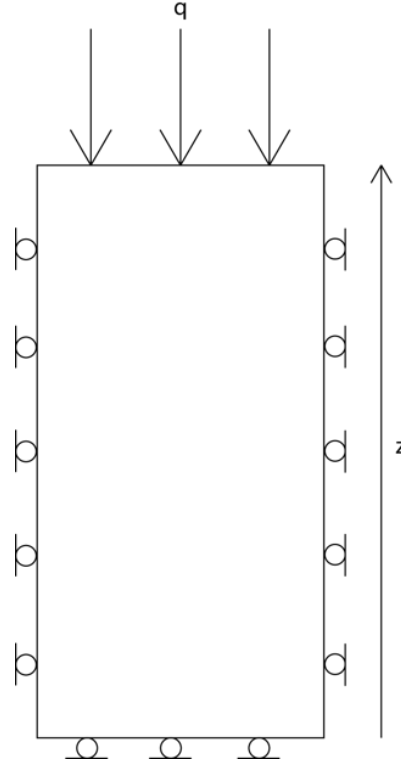


Figure 19: Terzaghi experiment setup

The significant parameters defined in my model are the drained Poisson's ratio, ν , the drained Young's Modulus, E , the bulk modulus of the grains, K_s , the initial porosity, ϕ , the fluid compressibility, c_f , the permeability, k , and the fluid viscosity, μ_f .

To begin the analytical calculation for pore pressure from ?, first the compressibility of the solid material is calculated

$$C_s = \frac{1}{K_s} \quad (85)$$

along with the shear modulus,

$$G = \frac{E}{2(1 + \nu)}. \quad (86)$$

This is then used to calculate the drained bulk modulus,

$$K = \frac{2G(\nu + 1)}{3(1 - 2\nu)} \quad (87)$$

which is in turn used to calculate the compressibility of the porous medium

$$C_m = \frac{1}{K}. \quad (88)$$

Then, the Biot Coefficient can be calculated,

$$\alpha = 1 - \frac{C_s}{C_m}. \quad (89)$$

This allows for the calculation of the storativity,

$$S = \phi c_f + (\alpha - \phi) C_s \quad (90)$$

which can be used in combination with the confined compressibility of the porous medium,

$$m_v = \frac{1}{K + \frac{4}{3}G} \quad (91)$$

to calculate the consolidation coefficient,

$$c_v = \frac{k_f}{\gamma_f (S + \alpha^2 m_v)}. \quad (92)$$

At this point, the initial pressure increase seen everywhere in the sample at the application of the load can be calculated,

$$p_0 = \frac{\alpha m_v}{S + \alpha^2 m_v} q. \quad (93)$$

Finally, the dimensionless pressure throughout the sample at a given time and location can be found,

$$\frac{p}{p_0} = \frac{4}{\pi} \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{2k-1} \cos \left[(2k-1) \frac{\pi z}{2h} \right] \exp \left[- (2k-1)^2 \frac{\pi^2}{4} \frac{c_v t}{h^2} \right] \quad (94)$$

The comparison between this analytical solution and my simulator's results can be seen in Figure ?? for various times.

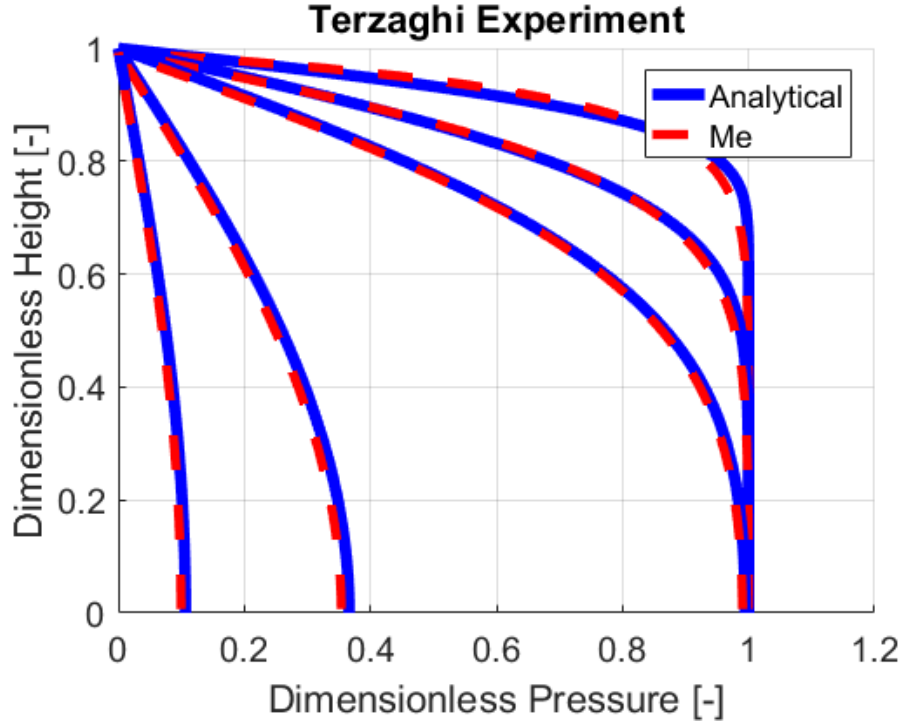


Figure 20: Comparison of P_p with an analytical solution for Terzaghi's problem

To calculate the displacements analytically, we will follow (?). To begin the Biot Modulus must be calculated using ?,

$$M = \left(\frac{\alpha - \phi_0}{K_s} + \frac{\phi}{K_f} \right)^{-1}. \quad (95)$$

Then, the Skempton coefficient can be calculated (?),

$$B = \frac{\alpha M}{K + \alpha^2 M} \quad (96)$$

This can be used to calculate the undrained Poissons ratio (?),

$$\nu_u = \frac{3\nu + \alpha B (1 - 2\nu)}{3 - \alpha B (1 - 2\nu)} \quad (97)$$

Together with the hydraulic conductivity ($k_h = \frac{k}{\mu_f}$), the Terzaghi consolidation coefficient can be calculated (?),

$$c = \frac{ck_h G (1 - \nu) (\nu_u - \nu)}{\alpha^2 (1 - 2\nu)^2 (1 - \nu_u)}. \quad (98)$$

Next, ? defines $z^* = \frac{z}{L}$ and $t^* = \frac{cT}{4L^2}$ before presenting the equation for the vertical displacement

$$u_z(z, t) = \frac{qL(1-2\nu_u)}{2G(1-\nu_u)}(1-z^*) + \frac{qL(\nu_u-\nu)}{2G(1-\nu_u)(1-\nu)}F_2(z^*, t^*), \quad (99)$$

where

$$F_2(z^*, t^*) = \sum_{m=1,3,\dots}^{\infty} \frac{8}{m^2\pi^2} \cos\left(\frac{m\pi z^*}{2}\right) (1 - \exp[-m^2\pi^2 t^*]). \quad (100)$$

The comparison of the analytical results and the results from my simulator can be seen in Figure ??.

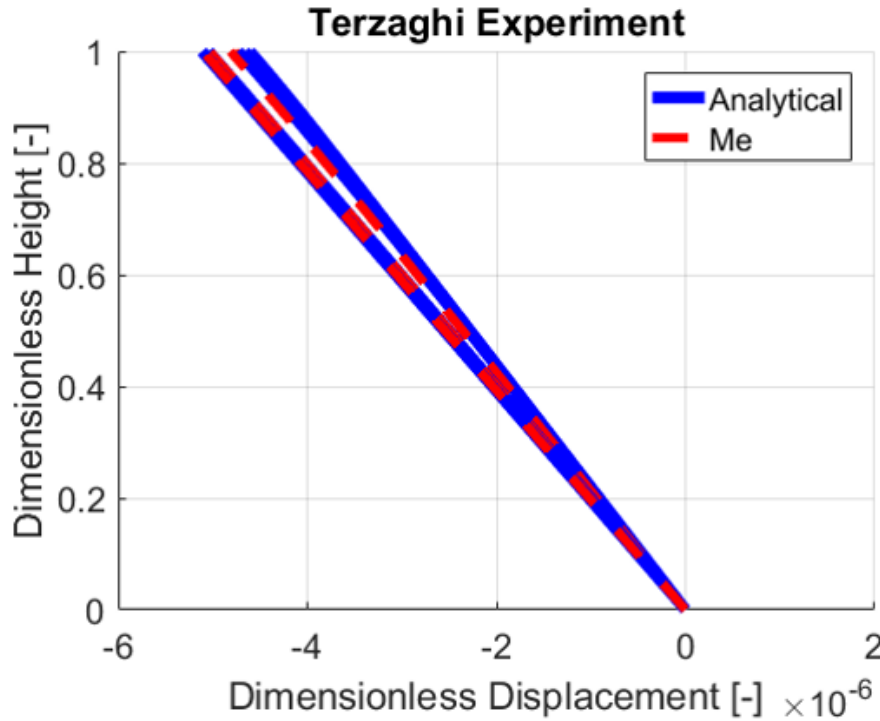


Figure 21: Comparison of u_z with an analytical solution for Terzaghi's problem

3.5 Poroelastic Validation - Mandel

Mandel's problem is a sample confined on two sides (top and bottom) by an impermeable membrane. The sides are left open and at atmospheric pressure. A force is applied on the top and bottom members which is equally distributed and equal to $2F$. This is a force per unit length in the z -direction (into the plane in this plane strain problem). Because of the symmetry of the problem, only the top right quadrant needs to be simulated. Then, the bottom and left boundaries are no-flow no-displacement. The fluid and grains are compressible. The schematic of the setup can be seen in Figure ??.

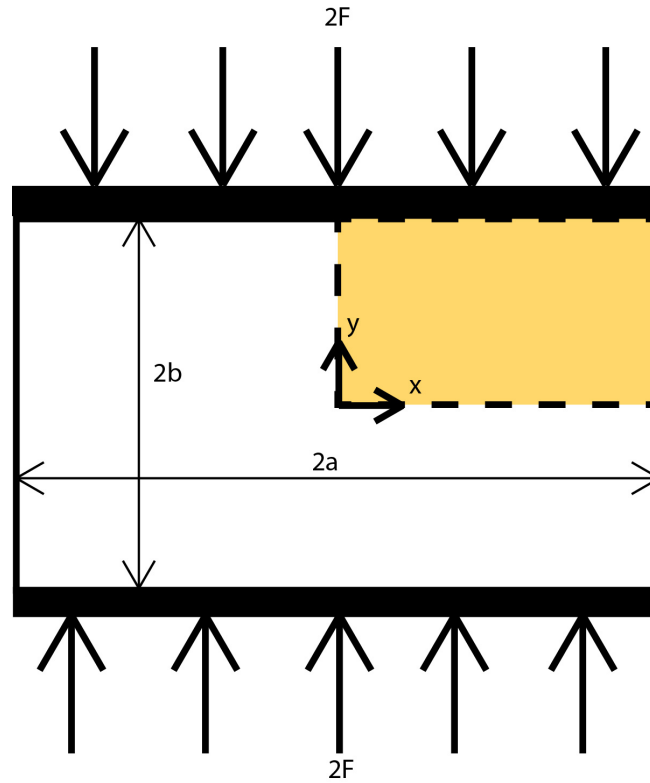


Figure 22: Mandel experiment setup

Here, a similar process is adopted as in Terzaghi's problem. First, the shear modulus, drained bulk modulus, Biot's coefficient, the fluid bulk modulus, Biot's modulus, and the Skempton coefficient are all calculated in that order. Next, the undrained Poisson's ratio is calculated (?),

$$\nu_u = \frac{3\nu + \alpha B (1 - 2\nu)}{3 - \alpha B (1 - 2\nu)}. \quad (101)$$

Then, the uniaxial drained bulk modulus is found (?),

$$K_v = \frac{2G (1 - \nu)}{(1 - 2\nu)}. \quad (102)$$

Next, the fluid diffusivity coefficient is calculated (?),

$$c_v = k_h M \frac{K_v}{K_v + \alpha^2 M}. \quad (103)$$

Next, an auxiliary elastic constant is calculated using (?)

$$\xi = \frac{1 - \nu}{\nu_u - \nu}. \quad (104)$$

This is used to find all of the positive roots, β_n , that satisfy (?),

$$\tan(\beta_n) = \xi\beta_n \quad (105)$$

Dimensionless pore pressure can be found using (?)

$$\frac{p}{p_0} = 2 \sum_{n=1}^{\infty} \frac{\sin(\beta_n)}{\beta_n - \sin(\beta_n) \cos(\beta_n)} \left(\cos\left(\frac{\beta_n x}{a}\right) \cos(\beta_n) \right) \exp\left(-\frac{\beta_n^2 c_v t}{a^2}\right), \quad (106)$$

with (?)

$$p_0 = \frac{B}{3a} (1 + \nu_u) F \quad (107)$$

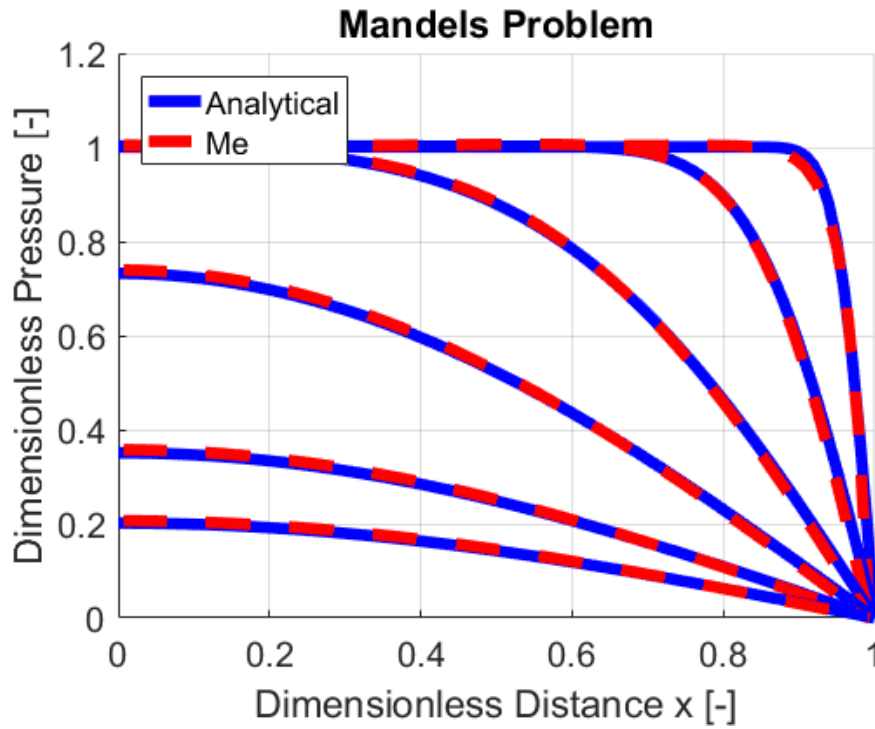


Figure 23: Comparison of pressure with an analytical solution for Mandel's problem

Next, the displacement in the y-direction can be calculated using (?),

$$u_y = \left[-\frac{F(1-\nu)}{2Ga} + \frac{F(1-\nu_u)}{Ga} \sum_{n=1}^{\infty} \frac{\sin(\beta_n) \cos(\beta_n)}{\beta_n - \sin(\beta_n) \cos(\beta_n)} \exp\left(-\frac{\beta_n^2 c_v t}{a^2}\right) \right] y \quad (108)$$

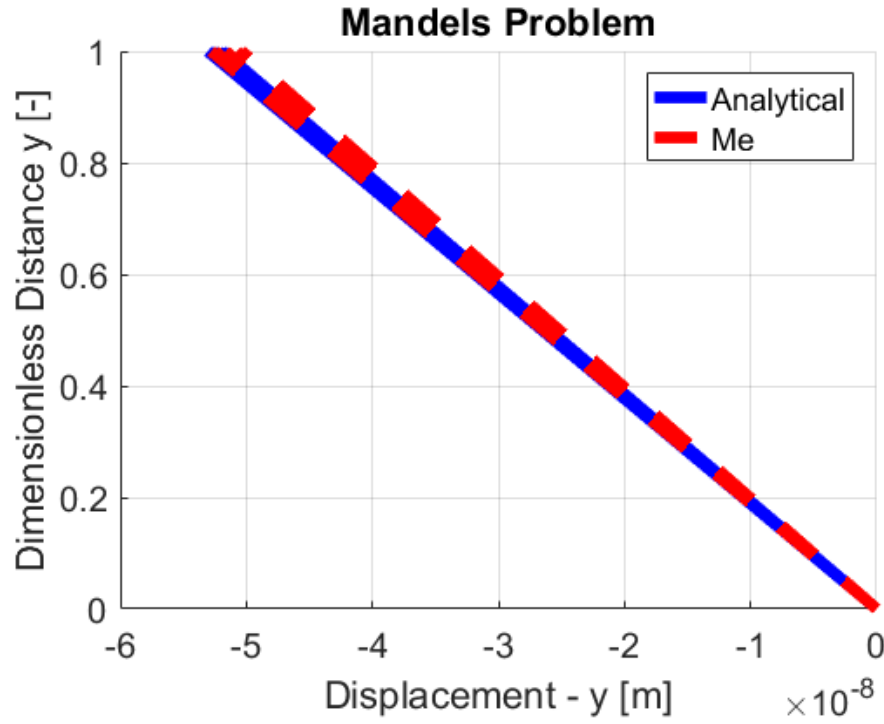


Figure 24: Comparison of u_y with an analytical solution for Mandel's problem

3.6 Poroelastic Validation - COMSOL

4 Appendix

4.1 Local Stiffness Matrix Terms

$$k_m(i, j) = \int_{x=0}^a \int_{y=0}^b (N^T B^T D B N) \{i, j\} \partial x \partial y \quad (109)$$

For plane stress:

$$A = \frac{E}{1 - \nu^2} \quad (110)$$

$$C = 1 \quad (111)$$

$$Q = \frac{1 - \nu}{2} \quad (112)$$

For plane strain:

$$A = \frac{E}{(1 + \nu)(1 - 2\nu)} \quad (113)$$

$$C = 1 - \nu \quad (114)$$

$$Q = \frac{1 - 2\nu}{2} \quad (115)$$

Row 1

$$k_m(1, 1) = A \left[C \left(\frac{b}{3a} \right) + Q \left(\frac{a}{3b} \right) \right] \quad (116)$$

$$k_m(1, 2) = A \left[\nu \left(\frac{1}{4} \right) + Q \left(\frac{1}{4} \right) \right] \quad (117)$$

$$k_m(1, 3) = A \left[C \left(\frac{-b}{3a} \right) + Q \left(\frac{a}{6b} \right) \right] \quad (118)$$

$$k_m(1, 4) = A \left[\nu \left(\frac{1}{4} \right) + Q \left(\frac{-1}{4} \right) \right] \quad (119)$$

$$k_m(1, 5) = A \left[C \left(\frac{b}{6a} \right) + Q \left(\frac{-a}{3b} \right) \right] \quad (120)$$

$$k_m(1, 6) = A \left[\nu \left(\frac{-1}{4} \right) + Q \left(\frac{1}{4} \right) \right] \quad (121)$$

$$k_m(1, 7) = A \left[C \left(\frac{-b}{6a} \right) + Q \left(\frac{-a}{6b} \right) \right] \quad (122)$$

$$k_m(1, 8) = A \left[\nu \left(\frac{-1}{4} \right) + Q \left(\frac{-1}{4} \right) \right] \quad (123)$$

Row 2

$$k_m(2, 1) = A \left[\nu \left(\frac{1}{4} \right) + Q \left(\frac{1}{4} \right) \right] \quad (124)$$

$$k_m(2, 2) = A \left[C \left(\frac{a}{3b} \right) + Q \left(\frac{b}{3a} \right) \right] \quad (125)$$

$$k_m(2, 3) = A \left[\nu \left(\frac{-1}{4} \right) + Q \left(\frac{1}{4} \right) \right] \quad (126)$$

$$k_m(2, 4) = A \left[C \left(\frac{a}{6b} \right) + Q \left(\frac{-b}{3a} \right) \right] \quad (127)$$

$$k_m(2, 5) = A \left[\nu \left(\frac{1}{4} \right) + Q \left(\frac{-1}{4} \right) \right] \quad (128)$$

$$k_m(2, 6) = A \left[C \left(\frac{-a}{3b} \right) + Q \left(\frac{b}{6a} \right) \right] \quad (129)$$

$$k_m(2, 7) = A \left[\nu \left(\frac{-1}{4} \right) + Q \left(\frac{-1}{4} \right) \right] \quad (130)$$

$$k_m(2, 8) = A \left[C \left(\frac{-a}{6b} \right) + Q \left(\frac{-b}{6a} \right) \right] \quad (131)$$

Row 3

$$k_m(3, 1) = A \left[C \left(\frac{-b}{3a} \right) + Q \left(\frac{a}{6b} \right) \right] \quad (132)$$

$$k_m(3, 2) = A \left[\nu \left(\frac{-1}{4} \right) + Q \left(\frac{1}{4} \right) \right] \quad (133)$$

$$k_m(3, 3) = A \left[C \left(\frac{b}{3a} \right) + Q \left(\frac{a}{3b} \right) \right] \quad (134)$$

$$k_m(3, 4) = A \left[\nu \left(\frac{-1}{4} \right) + Q \left(\frac{-1}{4} \right) \right] \quad (135)$$

$$k_m(3, 5) = A \left[C \left(\frac{-b}{6a} \right) + Q \left(\frac{-a}{6b} \right) \right] \quad (136)$$

$$k_m(3, 6) = A \left[\nu \left(\frac{1}{4} \right) + Q \left(\frac{1}{4} \right) \right] \quad (137)$$

$$k_m(3, 7) = A \left[C \left(\frac{b}{6a} \right) + Q \left(\frac{-a}{3b} \right) \right] \quad (138)$$

$$k_m(3, 8) = A \left[\nu \left(\frac{1}{4} \right) + Q \left(\frac{-1}{4} \right) \right] \quad (139)$$

Row 4

$$k_m(4, 1) = A \left[\nu \left(\frac{1}{4} \right) + Q \left(\frac{-1}{4} \right) \right] \quad (140)$$

$$k_m(4, 2) = A \left[C \left(\frac{a}{6b} \right) + Q \left(\frac{-b}{3a} \right) \right] \quad (141)$$

$$k_m(4, 3) = A \left[\nu \left(\frac{-1}{4} \right) + Q \left(\frac{-1}{4} \right) \right] \quad (142)$$

$$k_m(4, 4) = A \left[C \left(\frac{a}{3b} \right) + Q \left(\frac{b}{3a} \right) \right] \quad (143)$$

$$k_m(4, 5) = A \left[\nu \left(\frac{1}{4} \right) + Q \left(\frac{1}{4} \right) \right] \quad (144)$$

$$k_m(4, 6) = A \left[C \left(\frac{-a}{6b} \right) + Q \left(\frac{-b}{6a} \right) \right] \quad (145)$$

$$k_m(4, 7) = A \left[\nu \left(\frac{-1}{4} \right) + Q \left(\frac{1}{4} \right) \right] \quad (146)$$

$$k_m(4, 8) = A \left[C \left(\frac{-a}{3b} \right) + Q \left(\frac{b}{6a} \right) \right] \quad (147)$$

Row 5

$$k_m(5, 1) = A \left[C \left(\frac{b}{6a} \right) + Q \left(\frac{-a}{3b} \right) \right] \quad (148)$$

$$k_m(5, 2) = A \left[\nu \left(\frac{1}{4} \right) + Q \left(\frac{-1}{4} \right) \right] \quad (149)$$

$$k_m(5, 3) = A \left[C \left(\frac{-b}{6a} \right) + Q \left(\frac{-a}{6b} \right) \right] \quad (150)$$

$$k_m(5, 4) = A \left[\nu \left(\frac{1}{4} \right) + Q \left(\frac{1}{4} \right) \right] \quad (151)$$

$$k_m(5, 5) = A \left[C \left(\frac{b}{3a} \right) + Q \left(\frac{a}{3b} \right) \right] \quad (152)$$

$$k_m(5, 6) = A \left[\nu \left(\frac{-1}{4} \right) + Q \left(\frac{-1}{4} \right) \right] \quad (153)$$

$$k_m(5, 7) = A \left[C \left(\frac{-b}{3a} \right) + Q \left(\frac{a}{6b} \right) \right] \quad (154)$$

$$k_m(5, 8) = A \left[\nu \left(\frac{-1}{4} \right) + Q \left(\frac{1}{4} \right) \right] \quad (155)$$

Row 6

$$k_m(6, 1) = A \left[\nu \left(\frac{-1}{4} \right) + Q \left(\frac{1}{4} \right) \right] \quad (156)$$

$$k_m(6, 2) = A \left[C \left(\frac{-a}{3b} \right) + Q \left(\frac{b}{6a} \right) \right] \quad (157)$$

$$k_m(6, 3) = A \left[\nu \left(\frac{1}{4} \right) + Q \left(\frac{1}{4} \right) \right] \quad (158)$$

$$k_m(6, 4) = A \left[C \left(\frac{-a}{6b} \right) + Q \left(\frac{-b}{6a} \right) \right] \quad (159)$$

$$k_m(6, 5) = A \left[\nu \left(\frac{-1}{4} \right) + Q \left(\frac{-1}{4} \right) \right] \quad (160)$$

$$k_m(6, 6) = A \left[C \left(\frac{a}{3b} \right) + Q \left(\frac{b}{3a} \right) \right] \quad (161)$$

$$k_m(6, 7) = A \left[\nu \left(\frac{1}{4} \right) + Q \left(\frac{-1}{4} \right) \right] \quad (162)$$

$$k_m(6, 8) = A \left[C \left(\frac{a}{6b} \right) + Q \left(\frac{-b}{3a} \right) \right] \quad (163)$$

Row 7

$$k_m(7, 1) = A \left[C \left(\frac{-b}{6a} \right) + Q \left(\frac{-a}{6b} \right) \right] \quad (164)$$

$$k_m(7, 2) = A \left[\nu \left(\frac{-1}{4} \right) + Q \left(\frac{-1}{4} \right) \right] \quad (165)$$

$$k_m(7, 3) = A \left[C \left(\frac{b}{6a} \right) + Q \left(\frac{-a}{3b} \right) \right] \quad (166)$$

$$k_m(7, 4) = A \left[\nu \left(\frac{-1}{4} \right) + Q \left(\frac{1}{4} \right) \right] \quad (167)$$

$$k_m(7, 5) = A \left[C \left(\frac{-b}{3a} \right) + Q \left(\frac{a}{6b} \right) \right] \quad (168)$$

$$k_m(7, 6) = A \left[\nu \left(\frac{1}{4} \right) + Q \left(\frac{-1}{4} \right) \right] \quad (169)$$

$$k_m(7, 7) = A \left[C \left(\frac{b}{3a} \right) + Q \left(\frac{a}{3b} \right) \right] \quad (170)$$

$$k_m(7, 8) = A \left[\nu \left(\frac{1}{4} \right) + Q \left(\frac{1}{4} \right) \right] \quad (171)$$

Row 8

$$k_m(8, 1) = A \left[\nu \left(\frac{-1}{4} \right) + Q \left(\frac{-1}{4} \right) \right] \quad (172)$$

$$k_m(8, 2) = A \left[C \left(\frac{-a}{6b} \right) + Q \left(\frac{-b}{6a} \right) \right] \quad (173)$$

$$k_m(8, 3) = A \left[\nu \left(\frac{1}{4} \right) + Q \left(\frac{-1}{4} \right) \right] \quad (174)$$

$$k_m(8,4) = A \left[C \left(\frac{-a}{3b} \right) + Q \left(\frac{b}{6a} \right) \right] \quad (175)$$

$$k_m(8,5) = A \left[\nu \left(\frac{-1}{4} \right) + Q \left(\frac{1}{4} \right) \right] \quad (176)$$

$$k_m(8,6) = A \left[C \left(\frac{a}{6b} \right) + Q \left(\frac{-b}{3a} \right) \right] \quad (177)$$

$$k_m(8,7) = A \left[\nu \left(\frac{1}{4} \right) + Q \left(\frac{1}{4} \right) \right] \quad (178)$$

$$k_m(8,8) = A \left[C \left(\frac{a}{3b} \right) + Q \left(\frac{b}{3a} \right) \right] \quad (179)$$